

Waste-to-Filament: Building Circular Manufacturing Infrastructure in Bangladesh

1. Problem

Bangladesh generates approximately 821,000–1,000,000 tonnes of plastic waste annually, with only 30–37% recycled. Nearly two-thirds remain mismanaged, contributing to landfill overflow and environmental leakage. Simultaneously, Bangladesh relies almost entirely on imported 3D printing filament, priced at BDT 1,800–3,200 per kg (USD 16–30), creating high prototyping costs for students, startups, and SMEs. Plastic waste is abundant, while affordable manufacturing material is scarce.

2. Solution

We convert locally collected plastic waste into standardized, high-quality 3D printing filament, reducing import dependency while lowering costs for Bangladesh’s growing maker ecosystem.

3. Market Opportunity

3D printing adoption in Bangladesh is emerging but growing across universities, innovation labs, and SMEs. While precise national printer counts are unavailable, institutional adoption has expanded since the first university lab in 2014.

Estimated usage benchmarks (based on international averages): - Average desktop printer consumption: 7–20 kg/year - Typical institutional use: ~2 kg/month per active printer

Given import pricing of USD 16–30 per kg, even modest penetration into institutional markets represents a significant revenue opportunity.

Initial Focus (First 3 Years): - Target customers: Universities, innovation labs, prototyping SMEs - Projected annual sales target: 25–30 tonnes of filament

4. Business Model & Unit Economics

Revenue Streams: - Recycled filament sales (target price: USD 14–16 per kg) - Institutional supply contracts - Custom 3D printing services - Waste sourcing partnerships

Projected Cost Structure (per kg): - Collection & sorting: USD 2.50 - Processing & extrusion: USD 4.00 - Operations & distribution: USD 2.00 - Total estimated cost: USD 8.50

Projected Gross Margin: ~40–45%

5. Traction & Validation

- Lab-scale extrusion successfully tested
- Prototype filament produced and print-tested for dimensional consistency
- Initial interest from university innovation labs
- Global proof-of-concept demonstrated in comparable markets (India, Indonesia, Vietnam)

6. Competitive Advantage

- Access to abundant low-cost plastic waste feedstock
- Local production reduces import cost volatility
- 20–30% price reduction compared to imported filament
- Modular micro-processing hub model for regional scalability

7. Impact (Phase 1 – 3 Years)

- 25–30 tonnes of plastic diverted annually
- 20–35% reduction in prototyping costs for students and SMEs
- Reduced carbon footprint versus imported filament
- Supports circular economy transition in emerging manufacturing markets

8. Scalability Vision

Following validation in Bangladesh, the modular waste-to-filament micro-hub model can be replicated across emerging economies with high plastic waste and growing 3D printing adoption. Comparable markets such as India, Indonesia, and Vietnam demonstrate both demand growth and feasibility of recycled filament production.

9. Team

- Materials & polymer processing lead
- Waste operations & supply chain coordinator
- Business strategy & market development lead

We transform unmanaged plastic waste into affordable manufacturing input — building circular innovation infrastructure for emerging economies.